



Students' Knowledge and Attitude Towards Vaccination among Undergraduate Students at Iqra University, Islamabad Campus

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Abstract

Vaccination is one of the most effective methods of preventing infectious diseases, but vaccine hesitancy remains a major public health challenge due to misinformation and lack of awareness.

This study assessed the knowledge and attitudes of undergraduate students at Iqra University, Islamabad Campus, toward vaccination. A cross-sectional survey of 50 students was conducted using a structured questionnaire in MS Excel. Data were analyzed using descriptive statistics, Pivot Tables, and Pivot Charts. The findings provide insights into students' awareness and perceptions of vaccination and can support future educational programs to improve vaccine acceptance.

Introduction

Vaccination is one of the most successful public health interventions, preventing millions of deaths each year and controlling diseases such as smallpox, polio, measles, and hepatitis B. Despite these achievements, vaccine hesitancy remains a major global and national concern due to misinformation, lack of confidence, and limited awareness. In Pakistan, low immunization coverage and misconceptions about vaccines continue to contribute to vaccine-preventable diseases. University students play an important role in influencing public attitudes toward vaccination, making it essential to assess their knowledge and perceptions. Therefore, this study was conducted at Iqra University, Islamabad Campus, using a cross-sectional survey to evaluate students' knowledge and attitudes toward vaccination.



Problem

Vaccine hesitancy among university students is an important but under-researched issue in Pakistan. Many students are exposed to misinformation and conflicting vaccine information through social media and other sources, which can influence their attitudes and willingness to get vaccinated. Limited data on students' knowledge and perceptions makes it difficult to develop effective awareness programs. Therefore, this study aims to assess the knowledge and attitudes of undergraduate students at Iqra University, Islamabad Campus, and provide baseline data to support future vaccination education and awareness initiatives.





Objectives

- **General Objective**

- **Specific Objectives**



General Objective

To assess the knowledge and attitudes of undergraduate students at Iqra University, Islamabad Campus, toward vaccination.

Specific Objectives

- To determine the level of knowledge students have about vaccines, their mechanisms, types, and recommended schedules.
- To evaluate students' attitudes (positive, negative, or neutral) toward routine and COVID-19 vaccination.
- To identify common misconceptions or myths held by students regarding vaccine safety and efficacy.
- To explore the sources of vaccine-related information most commonly accessed by students (e.g., social media, healthcare providers, family).
- To examine the relationship between academic discipline (health vs. non-health programs) and vaccination knowledge/attitude.



Methodology

Study Design

Study Population and Location

Sample Size

Data Collection Tool

Data Analysis

Presentation

Web Page Integration





Study Design

A cross-sectional survey design will be used. This design is appropriate for this study as it allows the collection of data at a single point in time from a defined population, enabling the assessment of prevalence and associations without requiring long-term follow-up. Cross-sectional studies are cost-effective, relatively quick to execute, and well-suited for descriptive research objectives.



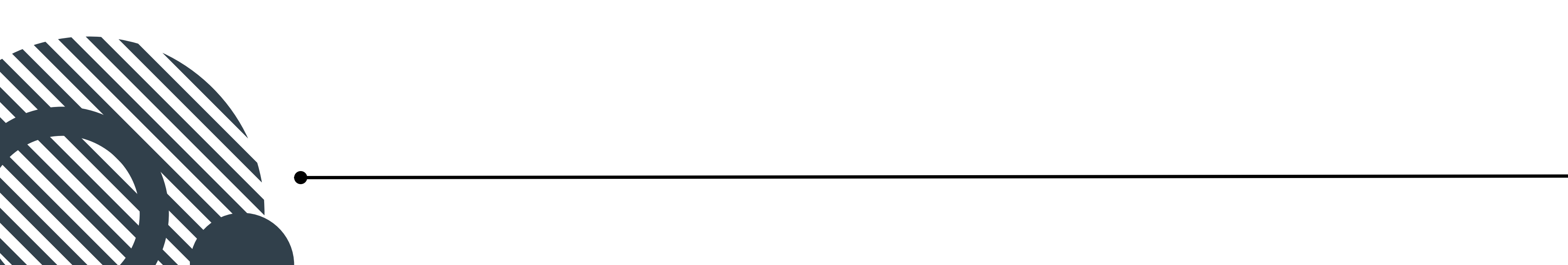


Component	Details
Study Population	Undergraduate students at Iqra University, Islamabad Campus
Study Location	Iqra University, Islamabad Campus
Sample Size	50 students
Sampling Method	Convenience sampling
Data Collection Tool	Structured questionnaire distributed via MS Excel / Google Forms
Data Analysis	Descriptive statistics using MS Excel (Pivot Tables & Pivot Chart Dashboards)
Presentation	Microsoft PowerPoint presentation
Dissemination	All work linked and published on a dedicated Web Page



Study Population and Location

The target population comprises undergraduate students enrolled at Iqra University, Islamabad Campus, across all departments and academic years. The study will be conducted on-campus during the current academic semester. Students who are willing to participate and capable of reading and understanding the questionnaire will be included. Students who are absent or unwilling to participate will be excluded.





Sample Size

A total of 50 students will be recruited for this study. Given the exploratory and descriptive nature of the research, this sample size is considered adequate for generating meaningful preliminary insights. Participants will be recruited using a convenience sampling technique, approaching students available on campus during data collection periods.





Data Collection Tool

A structured questionnaire will be the primary instrument for data collection. The questionnaire will be developed based on existing validated tools from the literature and will be customized to reflect the Pakistani socio-cultural context. It will be administered digitally through MS Excel (or Google Forms exported to Excel).



Questionnaire

The questionnaire will consist of the following sections:

- **Section A** – Demographic Information: age, gender, department, year of study.
- **Section B** – Knowledge Assessment: Multiple-choice and true/false questions assessing awareness of vaccine types, vaccine-preventable diseases, schedules, and mechanisms.
- **Section C** – Attitude Assessment: Likert-scale items (Strongly Agree to Strongly Disagree) evaluating perceptions of vaccine safety, necessity, effectiveness, and trust in health authorities.
- **Section D** – Sources of Information: Questions identifying where students primarily obtain vaccine-related information.

Questionnaire

Questionnaire

Date: _____

Location: _____ Interviewer: _____

Q.No	Question	Response Type
1	Are you a university student?	Yes / No
2	Have you received all routine vaccines?	Yes / No / Not sure
3	Are you fully vaccinated against COVID-19?	Yes / No / Partially
4	Do you believe vaccines are safe?	Strongly Agree / Agree / Disagree / Strongly Disagree
5	Main source of vaccine information	Social Media / Doctor / News / Family
6	Would you recommend vaccines to others?	Yes / No / Maybe
7	Reason for vaccine hesitancy (if any)	Side effects / Religious / Not needed

Ethical Consideration

Although this study does not involve clinical interventions, the following ethical guidelines will be observed:

- Informed consent will be obtained from all participants prior to survey administration.
 - Participation will be entirely voluntary, with no academic penalty for non-participation.
 - Anonymity and confidentiality of respondents will be strictly maintained; no personally identifiable information will be collected.
 - Data will be stored securely and used solely for academic research purposes.
-



Project Proposal



Data Analysis

All collected data will be entered and organized in Microsoft Excel. The following analytical approaches will be applied:

- Frequency distributions and percentages will be calculated for demographic variables and knowledge/attitude scores.
- Knowledge scores will be categorized as Poor ($< 50\%$), Moderate ($50-74\%$), or Good ($\geq 75\%$).
- Attitude scores will be classified as Negative, Neutral, or Positive based on aggregate Likert responses.
- Pivot Tables will be used to cross-tabulate variables (e.g., knowledge by department or gender).
- Pivot Chart Dashboards will be constructed to visually represent findings using bar charts, pie charts, and column graphs.

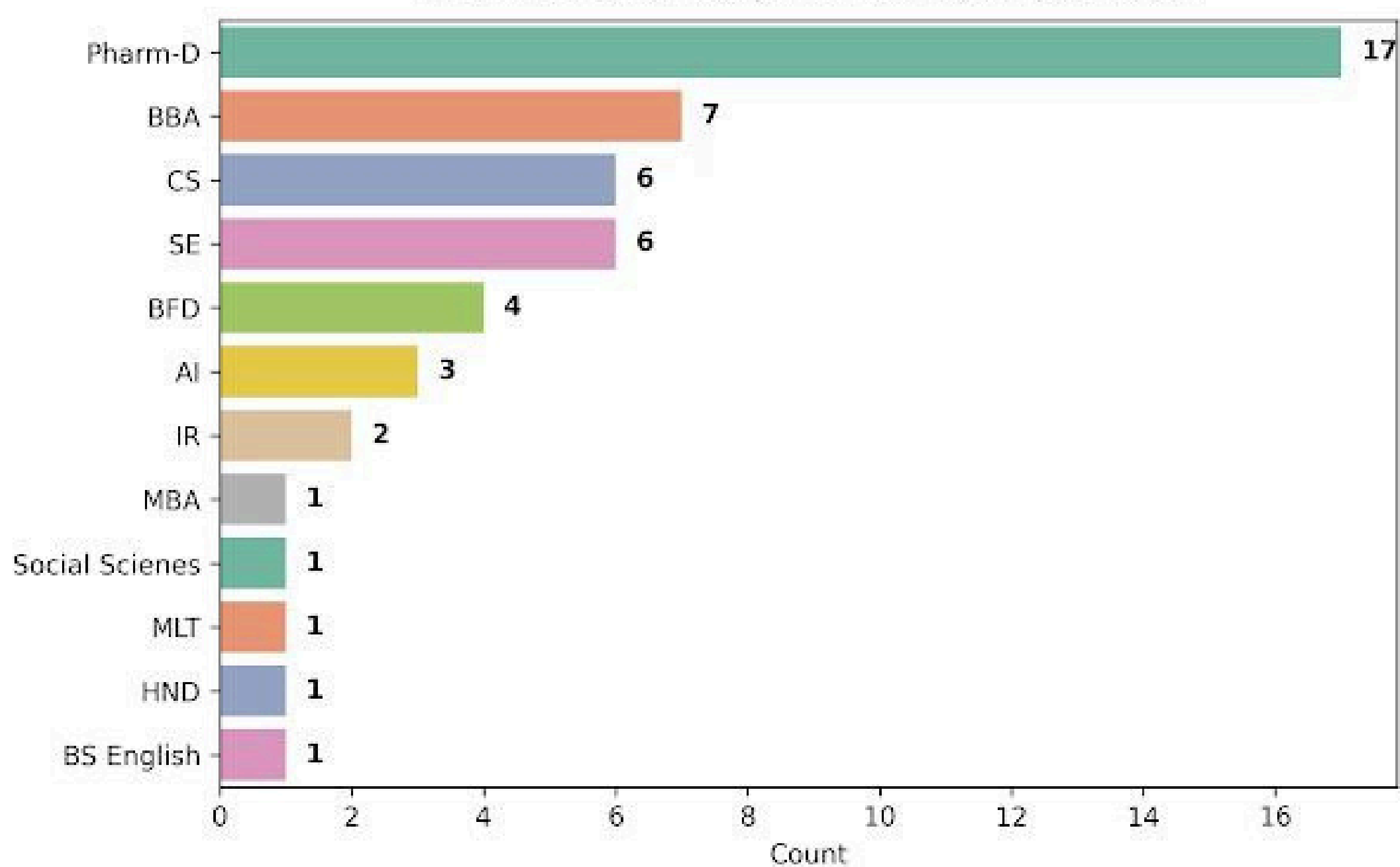


Result and Findings

1. Demographics & Academic Distribution

- The survey captures responses across 12 distinct academic departments.
- Pharm-D students make up the vast majority of respondents (34% or 17 out of 50).
- Business & IT majors (BBA, CS, SE, AI, MBA) represent a combined 46% (23 respondents).
- The remaining 20% is distributed across design (BFD), social sciences, international relations (IR), and other health sciences (MLT, HND).

Distribution of Respondents by Department

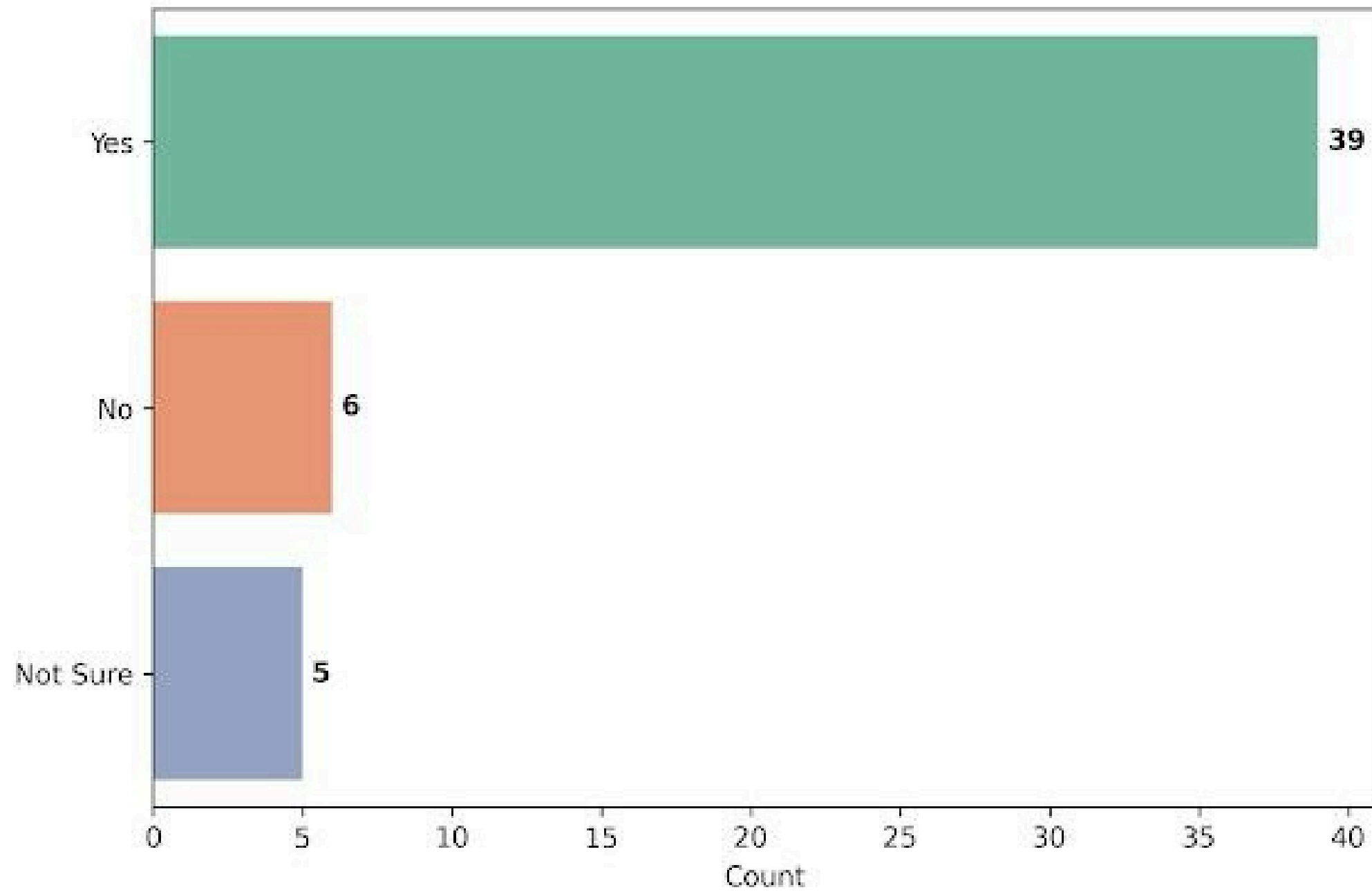


Result and Findings

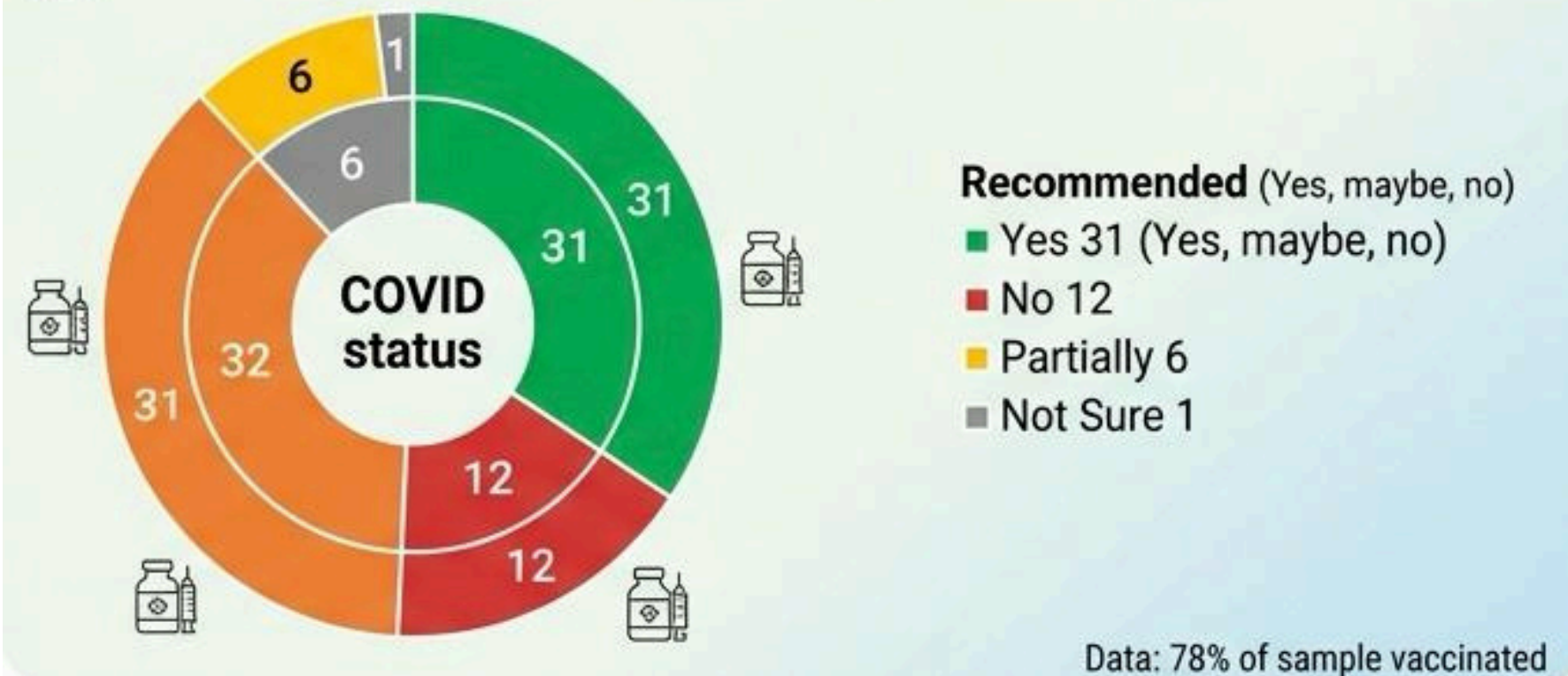
2. Vaccination Status & Acceptance

- Overall, the student body displays high trust and coverage when it comes to immunization.
- General Vaccination Acceptance: 78% of surveyed students explicitly state they are vaccinated in general, while 12% are not, and 10% remain unsure.
- COVID-19 Vaccination Breakdown:
 - Fully Vaccinated: 31 students (62%)
 - Partially Vaccinated: 6 students (12%)
 - Unvaccinated: 12 students (24%)
 - Unsure: 1 student (2%)
- Recommendation Rate: Peer-to-peer advocacy is incredibly strong, with 86% (43 students) actively recommending vaccines to others.

General Vaccination Status Breakdown



2 COVID-19 Vaccine Status vs. Recommended Vaccine



Result and Findings

3. Attitudes, Information Sources

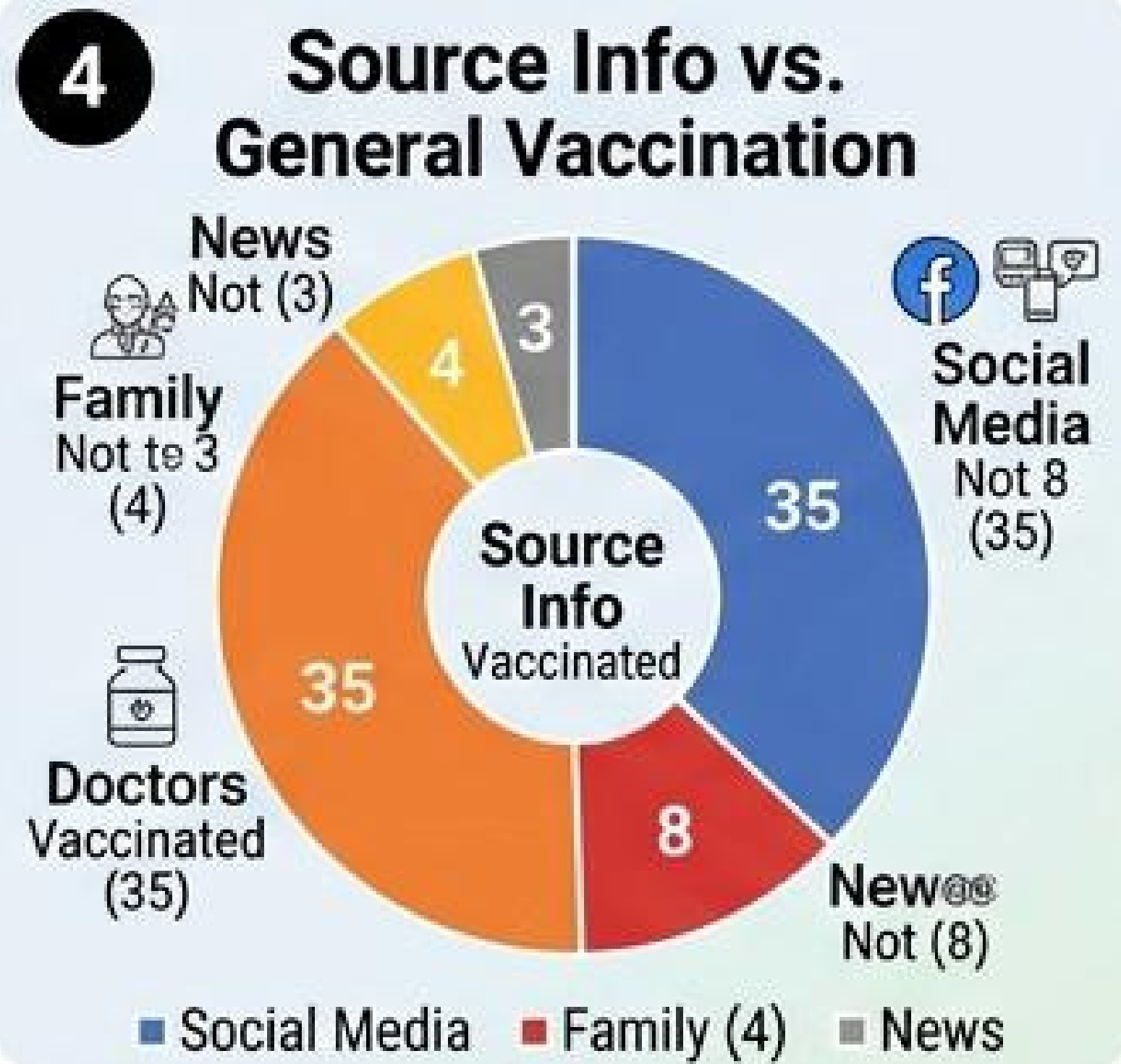
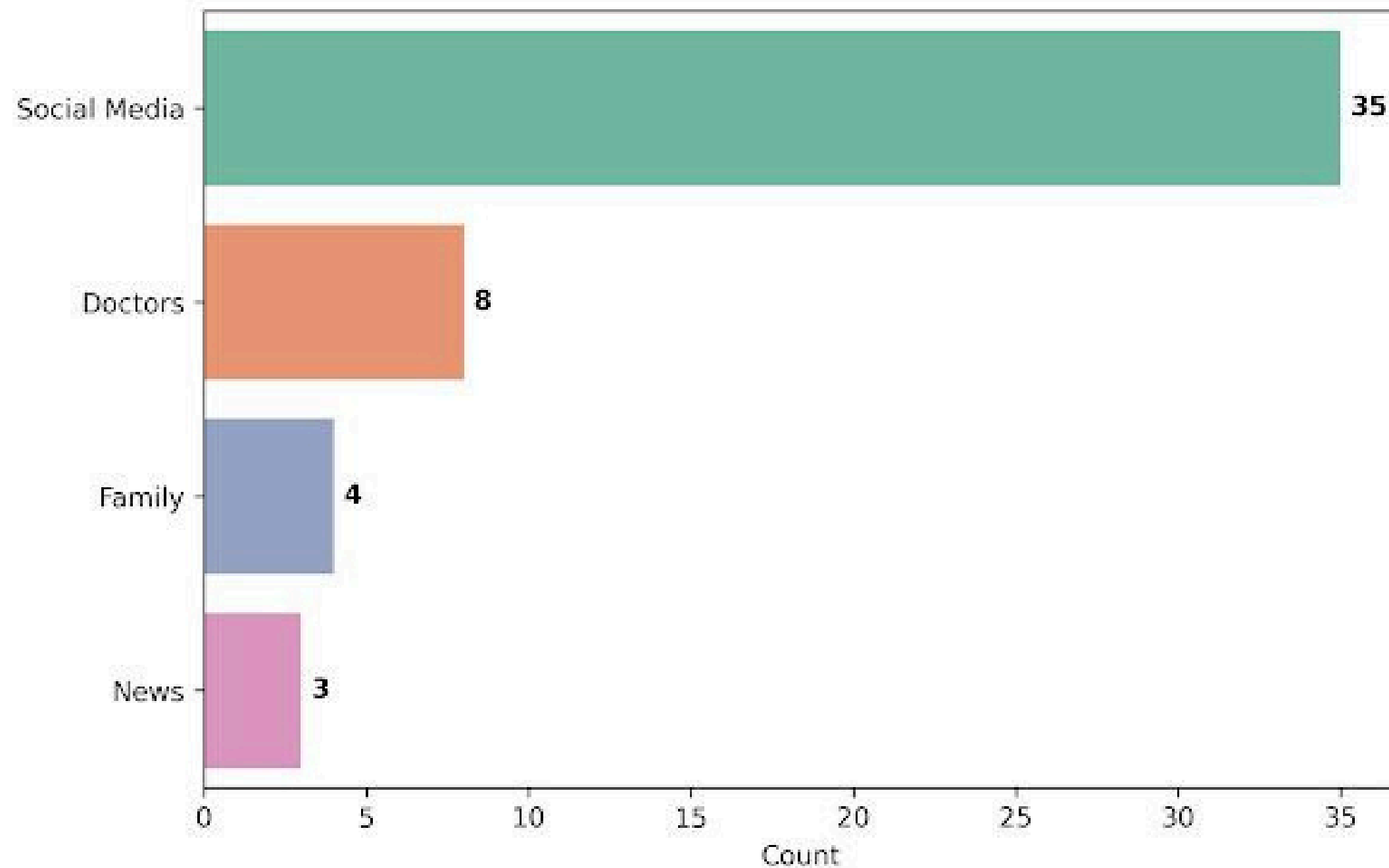
Vaccine Safety Beliefs:

- The overwhelming majority of students trust the medical safety of vaccines:
- 98% either Agree (27) or Strongly Agree (22) that vaccines are safe.
- Only 1 student (2%) expressed active disagreement regarding vaccine safety.

Main Sources of Information:

- Where students get their knowledge highlights a clear reliance on digital culture over traditional channels:
- Social Media: 70% (35 students) rely heavily on social platforms for their vaccine information.
- Medical Professionals: Combined, 16% (8 students) list "Doctor" or "Doctors" as their core source.
- Family & News Media: Combined to make up the remaining 14%.

Primary Source of Vaccine Information

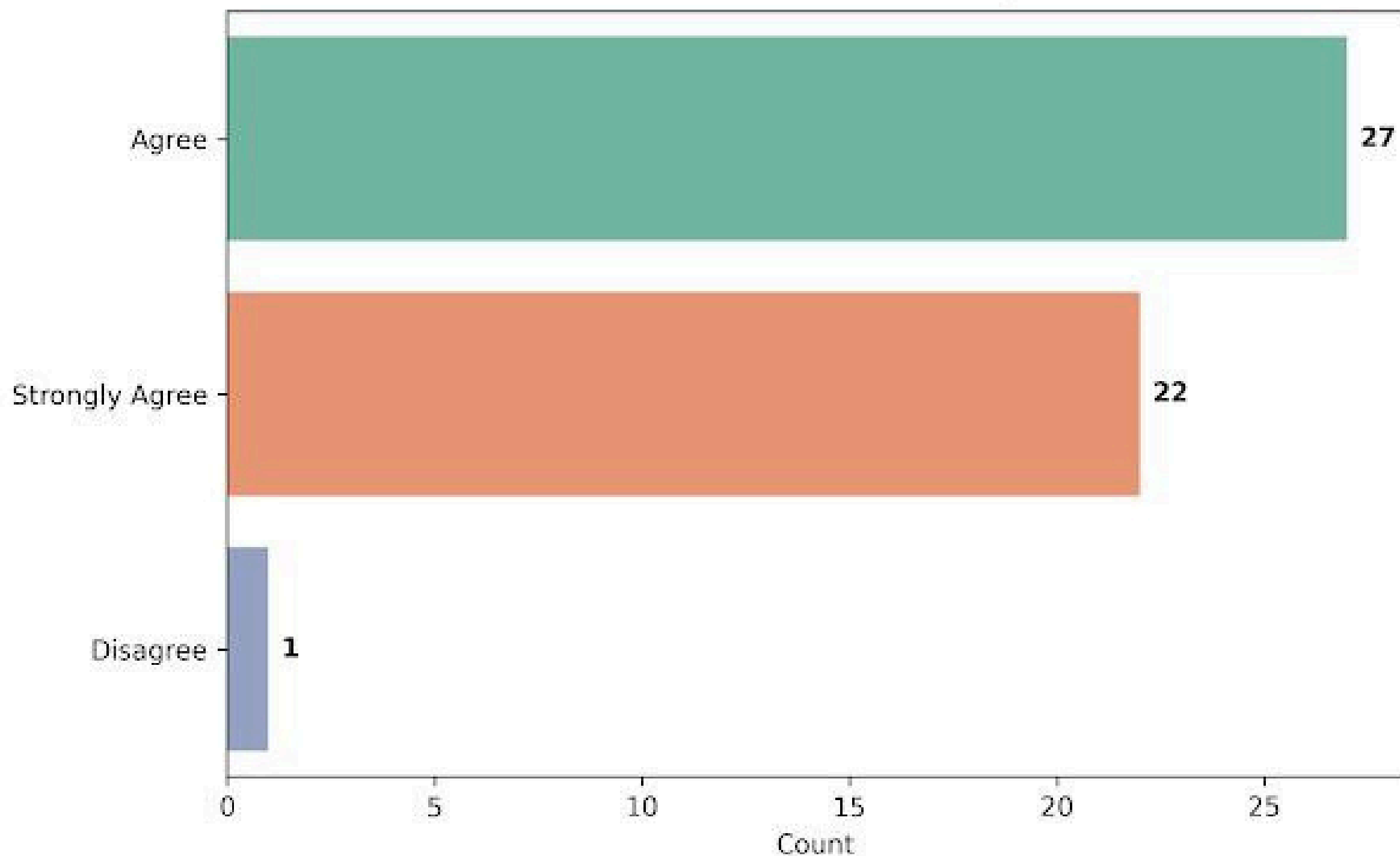


Result and Findings

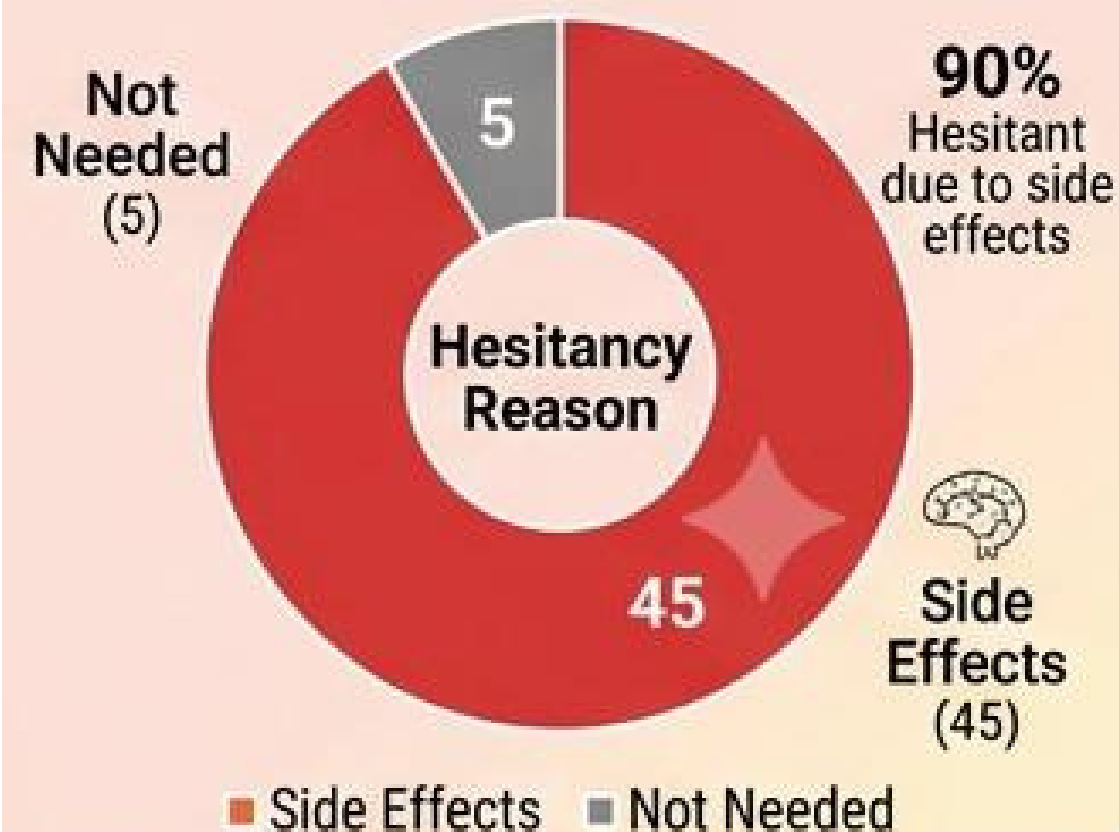
4. Key Drivers of Vaccine Hesitancy

- Despite high overall compliance, concerns still exist. When questioned about potential hesitancy factors:
- 90% (45 students) cited fear of Side Effects as their core concern.
- 10% (5 students) felt that the vaccines were simply Not Needed

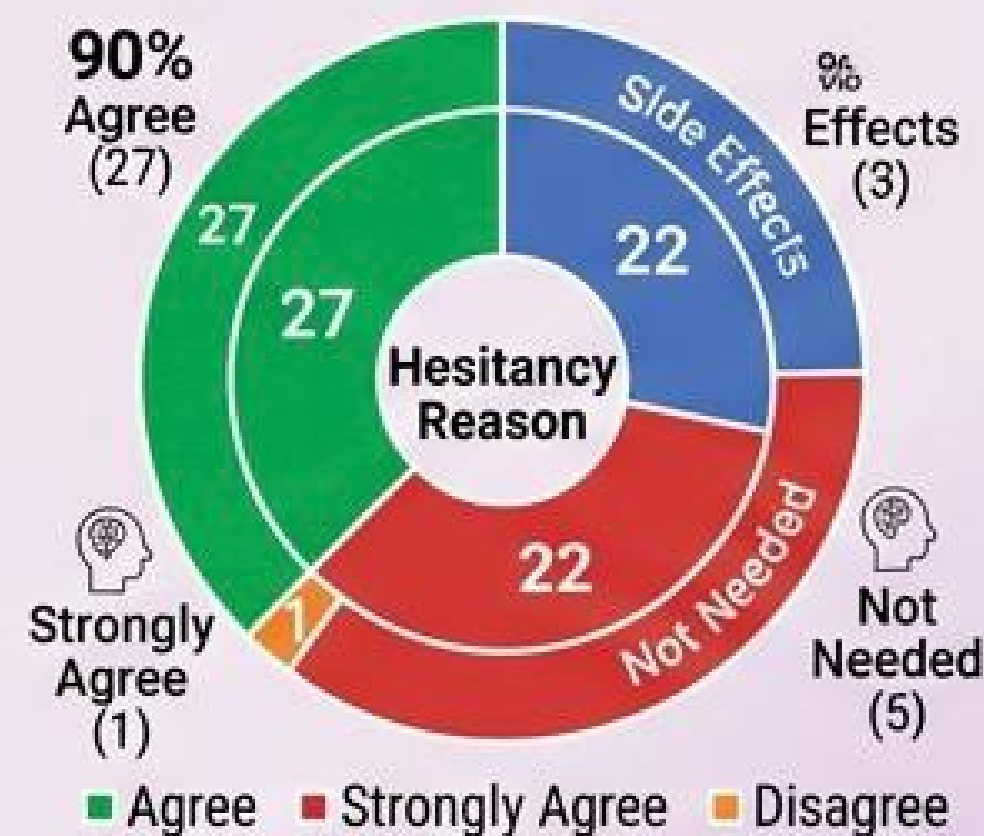
Belief in Vaccine Safety



6 Hesitancy Reason
(Total Breakdown)



3 Safety Belief vs. Hesitancy Reason





This study concludes that undergraduate students at Iqra University, Islamabad Campus, generally have a positive knowledge and attitude toward vaccination. Most participants were vaccinated, believed vaccines are safe and effective, and expressed willingness to recommend vaccination to others. However, fear of side effects and misinformation—particularly through social media—remain significant factors contributing to vaccine hesitancy.



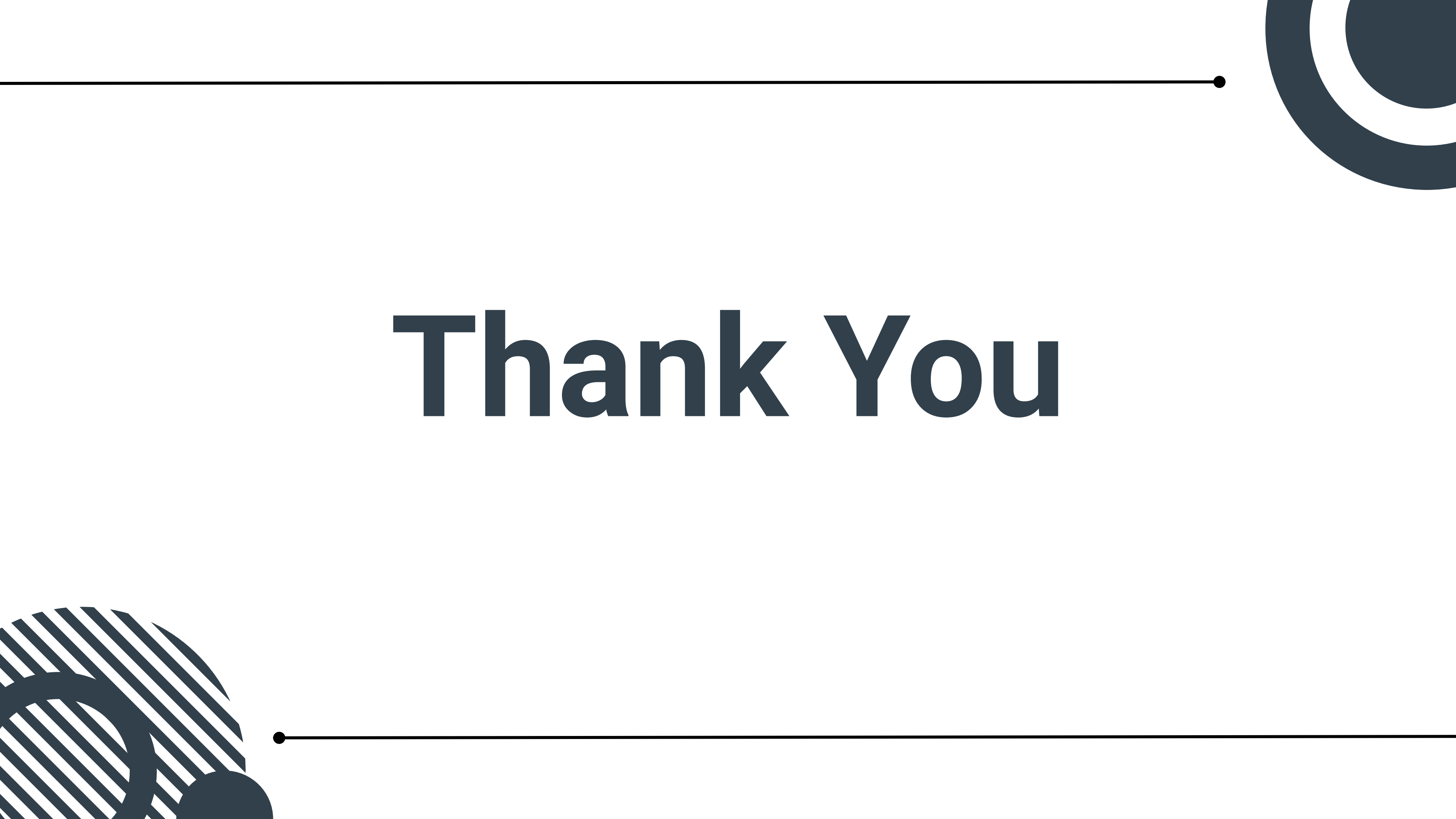
Conclusion

These findings highlight the importance of implementing targeted awareness campaigns, promoting reliable sources of health information, and strengthening vaccine education among university students to improve vaccine acceptance and support public health initiatives.



References

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Thank You
